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Oven

Abstract

An oven includes upper and lower assemblies, a fire box and a heat absorbing and emanating device. The upper assembly forms a cooking chamber and has a front side with an open entry to the cooking chamber. The lower assembly is located below the upper assembly and has an internal cavity. The fire box is located within the internal cavity and includes a heat source. The device is located between the upper and lower assemblies and absorbs heat from the heat source and emanates heat into the cooking chamber of the upper assembly. The upper assembly includes inner and outer dome assemblies. The inner dome is located within the outer dome and forms first and second airflow pathways.

Inventors: Bahreinian; Hessam (Newmarket, CA), Vandenharm; Robert (Woodbridge, CA)

Applicant: NewAge Products Inc. (Toronto, CA)

Family ID: 85385878

Assignee: NewAge Products Inc. (Toronto, CA)

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Primary Examiner: Pereiro; Jorge A

Attorney, Agent or Firm: Howard & Howard Attorneys PLLC

Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to ovens, and more specifically, to an oven having a lower assembly with a fire box, a cooking chamber and an upper assembly with two airflow pathways connected to the fire box.

BACKGROUND OF THE INVENTION

(2) Many people enjoy cooking, indoor or outdoor, to prepare or cook a variety of foods, for example, pizza. Ovens that are designed or adapted specifically for cooking pizzas are widely popular. Generally, such ovens are capable of reaching very high temperatures and cooking a pizza within minutes. However, traditional pizza ovens tend to be large, cumbersome and inefficient.

(3) The present invention is aimed at one or more of the problems identified above.

SUMMARY OF THE INVENTION

(4) In one aspect of the present invention, an oven having upper and lower assemblies, a firebox and a heat absorbing and emanating device is provided. The upper assembly forms a cooking chamber and has a front side with an open entry to the cooking chamber. The lower assembly is located below the upper assembly and has an internal cavity. The firebox is located within the internal cavity of the lower assembly and includes a heat source. The heat absorbing and emanating device is located between the upper and lower assemblies and is configured to absorb heat from the heat source and to emanate heat into the cooking chamber of the upper assembly. The upper assembly includes outer and inner dome assemblies. The inner dome is located within the outer dome and forms first and second airflow pathways, a second airflow pathway, each of the first and second airflow pathways having an open end located at a bottom of the upper assembly and being open to the internal cavity of the lower assembly, the first airflow pathway having a second end terminating at a heating vent located on an inner wall of the inner dome and configured to allow heated air from the heat source to enter the cooking chamber, the second airflow pathway having a second end located terminating in an upper portion of the upper assembly.

(5) In a second aspect of the present invention, an oven having upper and lower assemblies, a firebox, first and second heat sources, a heat guard, a thermometer, at least one removable door and heat absorbing and emanating device. The upper assembly has an upper portion and a bottom. The upper assembly forms a cooking chamber and has a front side with an open entry to the cooking chamber. The lower assembly is located below the upper assembly with the lower assembly having an internal cavity. The firebox is located within the internal cavity of the lower assembly. The first heat source is located within the firebox. The second heat source is located within the upper assembly for heating the cooking chamber. The heat guard is located between the second heat source and the cooking chamber. The thermometer is coupled to the lower assembly for sensing a temperature associated with the cooking chamber. The at least one removable door is rotatably coupled to the upper assembly within the open entry. The heat absorbing and emanating device is located between the upper and lower assemblies and is configured to absorb heat from the heat source and emanate heat into the cooking chamber of the upper assembly. The upper assembly includes an outer dome and an inner dome with the inner dome being located within the outer dome. First and second airflow pathways are disposed between the inner and outer domes. Each of the first and second airflow pathways has an open end located at the bottom of the upper assembly and are open to the internal cavity of the lower assembly. The first airflow pathway has a second end terminating at a heating vent located on an inner wall of the inner dome and is configured to allow heated air from the heat source to enter the cooking chamber. The second airflow pathway has a second end terminating in the upper portion of the upper assembly. The upper assembly further includes at least one exhaust vent located within the upper portion of the upper assembly near the second end of the second airflow path for allowing heated gas to exit the oven. The outer dome forms an exterior of the upper assembly. The inner dome forms at least a portion of the cooking chamber. The first and second airflow pathways being located between the inner and outer domes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Other advantages of the present invention will be readily appreciated as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings wherein:
- (2) FIG. 1A is a perspective view of an oven in a tabletop configuration, having an upper assembly and a lower assembly, according to a first embodiment of the present invention;
- (3) FIG. 1B is a front elevational view of the oven shown in FIG. 1A;
- (4) FIG. 1C is a perspective view of the oven shown in FIG. 1A in an elevated configuration;
- (5) FIG. 1D is a perspective view of the oven shown in FIG. 1A in a built-in configuration;
- (6) FIG. 1E is a perspective view of an enclosure that may be used with the oven shown in FIG. 1A;
- (7) FIG. 1F is a perspective view of a thermometer of the oven of FIG. 1A;
- (8) FIG. 2A is a cross-sectional view of the oven shown in FIG. 1A;
- (9) FIG. 2B is a cross-sectional view of the upper assembly of the oven shown in FIG. 1A;
- (10) FIG. 2C is a cross-sectional view of the lower assembly of the oven shown in FIG. 1A;
- (11) FIG. 3A is a first perspective view of an outer dome of the upper assembly of the oven of FIG. 1A;
- (12) FIG. 3B is a second perspective view of the outer dome of FIG. 3A;
- (13) FIG. 3C is a first perspective view of an outer dome top panel of the outer dome of FIG. 3A;
- (14) FIG. 3D is a second perspective view of the outer dome top panel of FIG. 3C;
- (15) FIG. 3E is a third perspective view of the outer dome top panel of FIG. 3C;
- (16) FIG. 3F is a first perspective view of an outer dome front panel of the outer dome of FIG. 3A;
- (17) FIG. 3G is a second perspective view of the outer dome front panel of FIG. 3F;
- (18) FIG. 3H is a perspective view of an outer dome back panel of the outer dome of FIG. 3A;
- (19) FIG. 3I is a perspective view of an outer dome side panel of the outer dome of FIG. 3A;
- (20) FIG. 4A is a first perspective view of a first door of the oven of FIG. 1A;
- (21) FIG. 4B is a second perspective view of the first door of FIG. 4A;
- (22) FIG. 4C is a first perspective view of an outer door panel of the first door of FIG. 4A;
- (23) FIG. 4D is a second perspective view of the outer door panel of FIG. 4C;
- (24) FIG. 4E is a perspective view of a plurality of door brackets of the first door of FIG. 4A;
- (25) FIG. 4F is a perspective view of a plurality of insulation panels of the first door of FIG. 4A;
- (26) FIG. 4G is a perspective view of a hinge for the first door;
- (27) FIG. 5A is a first perspective view of an inner dome, having an inner dome upper portion and an inner dome lower portion, according to an embodiment of the present invention;
- (28) FIG. 5B is a second perspective view of the inner dome of FIG. 5A;
- (29) FIG. 5C is a first perspective view of the inner dome lower portion of FIG. 5A;
- (30) FIG. 5D is a second perspective view of the inner dome lower portion of FIG. 5A;
- (31) FIG. 5E is a first perspective view of the inner dome upper portion of FIG. 5A;
- (32) FIG. 5F is a second perspective view of the inner dome upper portion of FIG. 5A;
- (33) FIG. 6A is a first perspective view of the lower assembly of the oven of FIG. 1A;
- (34) FIG. 6B is a second perspective view of the lower assembly of the oven of FIG. 1A;
- (35) FIG. 6C is a third perspective view of the lower assembly of the oven of FIG. 1A;
- (36) FIG. 7A is a first perspective view of an outer enclosure of the lower assembly of the oven of FIG. 1A;
- (37) FIG. 7B is a second perspective view of the outer enclosure of FIG. 7A;
- (38) FIG. 7C is a perspective view of a first side panel of the outer enclosure of FIG. 7A;
- (39) FIG. 7D is a perspective view of a second side panel of the outer enclosure of FIG. 7A;
- (40) FIG. 7E is a perspective view of a back panel of the outer enclosure of FIG. 7A;
- (41) FIG. 7F is a perspective view of a transformer assembly of the outer enclosure of FIG. 7A;

- (42) FIG. 8A is a first perspective view of an outer enclosure of the lower assembly of the oven of FIG. 1A;
- (43) FIG. 8B is a second perspective view of the outer enclosure of FIG. 8A;
- (44) FIG. 8C is a third perspective view of the outer enclosure of FIG. 8A;
- (45) FIG. 8D is a perspective view of a heat shield bottom panel of a heat shield of the outer enclosure of FIG. 8A;
- (46) FIG. 8E is a perspective view of a heat shield insulation panel of the heat shield;
- (47) FIG. 8F is a perspective view of a heat shield bottom panel of the heat shield;
- (48) FIG. 8G is a first perspective view of a fire box support structure of the outer assembly of FIG. 8A;
- (49) FIG. 8H is a second perspective of the fire box support structure of FIG. 8G;
- (50) FIG. 9A is a perspective view of a fire box of the oven of FIG. 1A;
- (51) FIG. 9B is a perspective view of a fire box back panel of the fire box of the FIG. 9A;
- (52) FIG. 9C is a perspective view of a fire box front panel of the fire box of FIG. 9A;
- (53) FIG. 9D is a perspective view of a first fire box side panel of the fire box of the FIG. 9A;
- (54) FIG. 9E is a perspective view of a second fire box side panel of the fire box of FIG. 9A;
- (55) FIG. 10 is a perspective view of a main burner of the oven of FIG. 1A;
- (56) FIG. 11 is a perspective view of a rear burner of the oven of FIG. 1A;
- (57) FIG. 12 is a perspective view of a control panel assembly of the oven of FIG. 1A; and,
- (58) FIG. 13 is a perspective view of a drawer assembly of the oven of FIG. 1A.

DETAILED DESCRIPTION OF INVENTION

(59) Referring to the FIGS., wherein like numerals indicate like or corresponding parts throughout the several views, an embodiment of an oven **10** having an upper assembly **100** and a lower assembly **200** for cooking items of food, such as pizza, as shown. With reference to FIGS. 2A-2C, the upper assembly **100** includes a cooking chamber **102** and an oven opening **104** through which the cooking chamber **102** may accessed.

(60) Returning to FIGS. 1A-1D, in the illustrated embodiment, the oven **10** includes first and second doors **12A**, **12B**. As explained in further detail below, the first and second doors **12A**, **12B** may commercial, dual-layered doors and include a sealing gasket (see below) and are rotatably coupled to the upper assembly **100** via respectively hinges **14**. The first and second doors **12A**, **12B** assist in quickly heating and maintaining the heat within the cooking chamber **102**. In another aspect of the present invention, the first and second doors **12A**, **12B** may be removable. The oven **10** includes first and second handles **16** mounted to the respective first and second doors **12A**, **12B**. In one embodiment, the handles **16** may be rotated to lock/unlock the doors. In the illustrated embodiment, the first and second doors **12A**, **12B** are locked when the handles **16** are upright and parallel (as shown in FIG. 1B). The first and second doors **12A**, **12B** may be unlocked by rotating the handles **16** in opposite directions, as indicated by the arrows in FIG. 1B. A thermometer **18** measures and displays the temperature of the cooking chamber **102**. The handles **16** may have a textured surface. For example, in the illustrated embodiment the surface of the handles **16** are knurled, i.e., have small ridges.

(61) The oven **10** may be provided or arranged in difference configurations. For example as shown in FIGS. 1A and 1B, the oven **10** may be arranged in a tabletop configuration in which the lower assembly **200** of oven **10** is mounted inside an enclosure **116** and a lower edge or bottom of the lower assembly **200** is suspended a short distance over a resting surface, such as a non-combustible countertop (not shown). With particular reference to FIG. 1E, the enclosure **116** may include a first enclosure side panel **116A**, a second enclosure side panel **116B** and a back enclosure panel **116C** which connects that first and second enclosure side panels **116A**, **116B**. Fuel or gas may be fed to the oven **10** via a swiveling elbow connector **116D**. A plurality of swivel leveling mounts **116E** are mounted to a bottom side of the first and second enclosure panels **116A**, **116B**. The swivel leveling mounts **116E** are adjustable and allow the oven **10** to be leveled.

(62) As shown, in FIG. 1C, the oven **10** may be arranged in an elevated configuration in which the lower assembly **200** is mounted within the enclosure **116** at an elevated or raised position. In the elevated configuration, the lower assembly **200** is mounted within the enclosure **116** such that the lower edge or bottom of the lower assembly **200** is elevated a greater distance above the resting surface (not shown) to provide additional airflow underneath the oven **10**.

(63) Additionally, the oven **10** may be provided or used without the enclosure **116** in a built-in configuration. The oven **10** in the built-in configuration may be mounted, for example, within other or existing indoor or outdoor cabinetry (not shown).

(64) As shown in FIGS. 2A-2C, the lower assembly **200** includes an internal cavity **202** and a fire box **204** located within the internal cavity **202**. The fire box **204** includes a first heat source **206**. In one embodiment, the first heat source **206** includes a first (or main) gas burner **206A**. In the illustrated embodiment, the first gas burner **206A** is “H-shaped” and is rated at **18,000 BTU**. A second (or back) heat source gas burner **206B** (see below) extends above the lower assembly **200** into the cooking chamber **102** of the upper assembly **100**. The second heat source **206B** may be a rear gas tube burner rated at, e.g., **22,000 BTU**.

(65) The oven **10** further includes a first heat absorbing and emanating device **60** located between the upper and lower assemblies **100**, **200**. The first heat absorbing and emanating device **60** is configured to absorb heat from the heat source **206** and emanate heat into the cooking chamber **102** of the upper assembly **100**. As shown in the illustrated embodiment, the first heat absorbing and emanating device **60** includes a plurality, e.g., three, of food grade ceramic cooking stones **62** that form a cooking surface **64**. The ceramic cooking stones **62** may have a thickness of 30 mm. The plurality of ceramic cooking stones **62** are heated directed from below by the first gas burner **206A** (as shown by arrows A) and emanate heat into the cooking chamber **102** to ensure that the bottom of the food product placed on the ceramic cooking stones **62** are heated thoroughly. The second gas burner **206B** directly heats the cooking chamber **102** to cook the food product. A heat guard **66** separates the second gas burner **206B** from the rest of the cooking chamber **102** of the upper assembly **100** to protect the food product from the flames from the second gas burner **206B**.

(66) In the illustrated embodiment, the oven **10** includes a second heat absorbing and emanating device **70** positioned at the top of the cooking chamber **102**. As shown, includes a ceramic cooking stone **72**.

(67) As shown in FIGS. 2A-2C, the upper assembly **100** includes an outer dome **106** and an inner dome **108**. The outer dome **106** forms the exterior of the upper assembly **100**. The inner dome **108** is located within the outer dome **106** and forms a first airflow pathway **110** and a second airflow pathway **112**. As shown, in the illustrated embodiment, each of the first and second airflow pathways **110**, **112** having a first end **110A**, **112A** located at a bottom of the upper assembly **100** and being open to the internal cavity **202** of the lower assembly **200**. The first airflow pathway **110** has a second end **110B** that terminates at a heating vent **114** located on an inner wall **108A** of the inner dome **108**. The first airflow pathway **110** is configured to allow heated air from the heat source **206** to enter the cooking chamber **102**. The second airflow pathway **112** has a second end **112B** located at an upper portion of the upper assembly **100**. As discussed in more detail below, the outer dome **106** includes a number of upper and lower (or exhaust) vents **118J**, **118O** to allow heated air to escape the oven **10**. In one embodiment, as shown in FIG. 2A, the first and second airflow pathways **110**, **112** are separated by an interior wall **111**.

(68) In operation, the ceramic cooking stone **72** of the second heat absorbing and emanating device **70** absorbs and retains heat from the heated air of the first airflow pathway **110**.

(69) As shown in FIG. 1F, the thermometer **18** includes a dial **18A** that includes an indicator (not shown), such as a radial indicator, to provide a visual indication of the temperature in the cooking chamber **102**. The thermometer **18** further includes a stem **18B** extending from the back of the dial **18A**. The stem **18B** is threaded at an upper end **18C** located adjacent the dial **18A**. A sensing element (not shown) is located within a lower end **18D** of the stem **18B**. A positioning pin **18E**

assists orientating the thermometer **18** within the upper assembly **100**.

(70) With reference to FIGS. 3A-3B, the outer dome **106** includes an outer dome top panel **118**, an outer dome front panel **120**, an outer dome first side panel **122**, an outer dome second side panel **124** and an outer dome back panel **126**. In one embodiment of the present invention, the outer dome panels **118**, **120**, **122**, **124**, **126** are composed from stainless steel and are connected using fasteners, such as screws, to form the outer dome **106**.

(71) With respect to FIGS. 3C, 3D and 3E, in the illustrated embodiment, the outer dome top panel **118** has a generally rectangular outer shape formed by first, second, third and fourth lower sides **118A**, **118B**, **118C**, **118D**. A rectangular top surface **118E** forms the top of the outer dome top panel **118**. Upper edges of each of the first, second, third and fourth sides **118A**, **118B**, **118C**, **118D** are connected to outer sides of the rectangular top surface **118E** by respective first, second, third and fourth sloping side walls **118F**, **118G**, **118H**, **118I**.

(72) One or more of the sloping side walls **118F**, **118G**, **118H**, **118I** may include a plurality of upper vents **118J**. As shown in FIG. 3E, first, second, third and fourth flanges **118K**, **118L**, **118M**, **118N** extend inward from the respective first, second, third and fourth sides **118A**, **118B**, **118C**, **118D**. One or more of the flanges **118K**, **118L**, **118M**, **118N** may include a plurality of downwardly facing lower vents **118O**. Returning to FIG. 2, the upper vents **118J** and the lower vents **118O** are located at the end of the second airflow pathway **112**. The upper and lower vents **118J**, **118O** act as exhaust vents by providing a pathway for heated air to exit the oven **10**.

(73) With reference to FIGS. 3F and 3G, the outer dome front panel **120** includes an outer dome front inner side **120A**. The outer dome front inner side **120A** includes a door cutout **120B** for receiving the first and second doors **12A**, **12B**. In the illustrated embodiment, a plurality of flanges **120C** extend away from (inwardly or outwardly) the edges of the outer dome front inner side **120A**. The flanges **120C** may include one or more apertures for receiving fasteners (not shown) to connect the outer dome front panel **120** to the outer dome top panel **118**, the outer dome first and second side panels **122**, **124** and/or the lower assembly **200**. The hinge **14** may be mounted to hinge support brackets **120D** which are connected to the flange **120C** extending inwardly from the door cutout **120B**. An outer dome front panel support **120E** connected to the flange **120C** extending inwardly from the door cutout **120B** may provide additional support to the first and second doors **12A**, **12B** and the hinges **14**. The front inner panel **12D** may further include a thermometer cutout **120F** for receiving the thermometer **18**.

(74) With reference to FIG. 3H, the outer dome back panel **126** includes an outer dome back side **126A** and a plurality of outer dome back panel flanges **126B** extending away from the edge of the outer dome back side **126A**. The outer dome back panel flanges **126B** may include one or more apertures or slots **126C** for receiving fasteners (not shown) to connect the outer dome back panel **126** to the outer dome side panels **122**, **124** and/or the lower assembly **200**. In the illustrated embodiment, a plurality of rivet nuts **126D** may be coupled to a lower one of the outer dome back panel flanges **126B** for receiving appropriate fasteners (not shown).

(75) With reference to FIG. 3I, a perspective view of the outer dome second side panel **124** is shown. The outer dome first side panel **122** is similar. The outer dome second side panel **124** includes an outer dome side panel side **124A** and outer dome side panel extensions **124B**. A plurality of flanges **124C** extend from top and bottom edges of the outer dome side panel side **124A** and from the outer dome side panel extensions **124B**. The flanges **124C** may include apertures **124D** or tabs with apertures **124E** for receiving fasteners (not shown) to connect the outer dome side panel **124** to the outer dome top panel **118**, the first and second side panels **122**, **124**, and the lower assembly **200**.

(76) With reference to FIGS. 4A-4G, the first door **12A** includes an outer door panel **12C** and an inner door panel **12D**. As shown, the hinge **14** includes an upper hinge portion **14A** and a lower hinge portion **14B**. The upper hinge portion **14A** includes apertures **14C** for receiving fasteners (not shown) to couple the first door **12A** to an inner surface of the outer door panel **12C**. The lower

hinge portion **14B** includes a plurality of apertures **14D** for receiving fasteners (not shown) to couple the first door **12A** to the outer dome front panel **120**. With reference to FIGS. **4C** and **4D**, the inner door panel **12D** includes a flange **12E** that extends from the outer periphery of the inner door panel **12D** towards the outer door panel **12C** to form an inner door cavity **12F** between the outer and inner door panels **12C**, **12D**. A notch **12M** within the flange **12E** receives the lower hinge portion **14B**. A plurality of door brackets **12H** (FIG. **4E**) and a plurality of insulation panels **121** are located within the inner door cavity **12F**. A seal **12J** may be affixed to at least a portion of an outer surface of the flange **12E**. In the illustrated embodiment, the seal **12J** has a diameter of 0.5 inches. The insulation panels **121** and the seal **12J** help to retain heat within the cooking chamber **102**. The inner door panel **12D** and the flange **12E** include a plurality of apertures **12K** for receiving respective fasteners to connect the door brackets **12H**, the outer door panel **12C** and the hinge **14** to the inner door panel **12D**. The outer door panel **12C** includes a door handle aperture **12L** for receiving the handle **16**.

(77) The second door **12B** is similar, but in an opposite configuration, to the first door **12A**, and therefore not shown in greater detail.

(78) As mentioned above, the first and second doors **12A**, **12B** may be removable. The upper hinge portion **14A** allows the respective first or second door **12A**, **12B** to be opened and closed and is removable from the lower portion **12B** by lifting the first or second door **12A**, **12B** and the upper hinge portion **14A** then the first or second door **12A**, **12B** is in the open position.

(79) As shown in FIGS. **1A-4B**, each handle **16** includes a handle base **16A**, an insulation washer **16B** and a handle portion **16C**. The handle base **16A** includes a handle base first portion **16A-1** is fastened to the respective first or second door **12A**, **12B** and extends outward therefrom. The insulation washer **16B** is positioned between the handle base first portion **16A-1** and the outer door panel **12C**. The handle base **16A** includes a handle base second portion **16A-2** that is coupled to the handle base first portion **16A-1** and extends at a right angle therefrom as shown. In the illustrated embodiment, the handle portion **16C** is located at an opposite end of the handle base second portion **16A-2**. The handle portion **16C** may have a textured surface, such as having grooves or ridges, to provide a gripping surface.

(80) With reference to FIGS. **5A-5G**, the inner dome **108** includes an inner dome upper portion **128** and a lower dome portion **130**. A plurality of insulation pads **132** are affixed, for example, by adhesive, to an outer surface **128A** of the inner dome upper portion **128** and the lower dome portion **130** to assist in keeping heat within the cooking chamber **102**. A pair of halogen light assemblies **134** are mounted to the outside of the inner dome upper portion **128** which provide lighting to the cooking chamber **102**. A plurality of insulation support brackets **136** mounted to the inner dome upper portion **128** by fasteners help maintain the insulation panels **132** mounted to the inner dome upper portion **128** in place.

(81) With specific reference to FIGS. **5A-5D**, the inner dome lower portion **130** includes an inner dome lower portion front panel **130A**, inner dome lower portion back panel **130B** and first and second inner dome lower portion side panels **130C**, **130D**. The inner dome lower portion front, back and side panels **130A**, **130B**, **130C**, **130D** are composed from stainless steel held together using appropriate fasteners (not shown). Flanges **130E** extend from the inner dome lower portion side panels **130A**, **130B**, **130C**, **130D** include apertures for receiving appropriate fasteners to connect the inner dome lower portion **130** to the inner dome upper portion **128** and to the lower assembly **200**. The inner dome lower portion front panel **130A** includes a door cutout **130F** that is aligned with the door cutout **120B** in the outer dome front panel **120**. The inner dome lower portion side panels **130C**, **130D** form part of the inner wall of the inner dome **108A** (see above). In the illustrated each inner dome lower portion side panels **130C**, **130D** includes an exhaust vent **114**. A first airflow pathway **110** is located on each side of the upper assembly **100** terminating at the respective heating vent **114**.

(82) With specific reference to FIGS. **5A**, **5B**, **5E** and **5F**, the inner dome upper portion **128**

includes an upper portion top surface **128A**. In the illustrated embodiment, the inner dome upper portion **128** has a generally rectangular shape define by the upper portion top surface **128A** and an upper portion flange **128B**. A plurality of upper portion side panels **128C** extend from the upper portion top surface **128A** to the upper portion flange **128B**. The upper portion top surface **128A**, upper portion flange **128B** and upper portion side panels **128C** are composed from stainless steel held together using appropriate fasteners (not shown).

(83) The upper portion top surface **128A** includes a cooking stone opening **128D**. As shown in FIGS. **2A** and **2B**, the ceramic cooking stone **72** of the second heat absorbing and emanating device **70** rests on the inner dome upper portion **128** such that the ceramic cooking stone **72** is exposed to the cooking chamber **102**. An inner dome upper portion **128** is fastened to the upper portion top surface **128A** and retains the ceramic cooking stone **72** in place.

(84) The pair of halogen light assemblies **134** are mounted to opposites upper portion side panels **128C**. Light from the halogen light assemblies **134** enters the cooking chamber **102** through a pair of respective light openings **128E** in the upper portion side panels **128C**.

(85) An inner dome heat exhaust panel **140** is fastened above the inner dome stone cover **138**. Air from the cooking chamber **102** exhausts through an exhaust opening **128F** and is routed towards the lower and upper vents **118O**, **118J** by an inner dome heat exhaust panel **140** fastened to the outer dome top panel **118** above the inner dome stone cover **138**.

(86) With reference to FIGS. **1D-1E**, **6A-6B**, **12** and **13**, the oven **10** includes a control panel assembly **142** and a drawer assembly **144**. The control panel assembly **142** provides controls to allow the burners **206A**, **206B** to be ignited and adjusted to heat the cooking chamber **102** to a desired temperature (see below). The drawer assembly **144** includes a drawer **144A** in which food may be placed and kept warm (see below). In the illustrated embodiment the drawer **144A** may have a depth of 15" and a depth of 14". The temperature within the drawer assembly **144** is dependent upon the temperature within the cooking chamber **102**. In the illustrated embodiment, the ambient temperature within the drawer may range from 80-200 degrees Fahrenheit.

(87) The control panel assembly **142** includes first and second knob assemblies **20**, **22** and first and second ignition buttons **24**, **26**. Each burner knob assembly **20**, **22** includes a respective rotatable knob **20A**, **22A** and bezel **20B**, **22B**. The first burner knob assembly and first and second ignition buttons **24**, **26** may be used to ignite the first or main gas burner **206A** and the second burner knob assembly and first and second ignition buttons **24**, **26** may be used to ignite the second or rear gas burner **206B**. For example, to ignite the first gas burner **206A**, the first burner knob assembly **20** must be rotated to an ignition position (as indicated on the respective bezel **20B**, **22B**) and pushed inward (to allow gas to flow to the main gas burner **206A**). Then the first ignition button **24** may be pressed to ignite the gas. Once the gas burner **206A**, **206B** has been ignited, the burner knob assemblies **20A**, **20B** to the desired level. The control panel assembly **142** further includes a see-through slide indicator **28**. The see-through slide indicator **28** includes an indicator knob **28A**. Indicator knob **28A** is moveable between an open position and a closed position (shown in FIG. **1A**). When the indicator knob **28A** is in the open position, the interior of the lower assembly **200** is viewable such that visual configuration that the gas burner **206A** is lit may be seen through a viewing glass (not shown).

(88) With reference to FIG. **13**, the drawer assembly **144** includes the drawer **144A**, a drawer front panel **144B**, a drawer handle **144C** and a warming rack **144D**. The warming rack **144D** fits within, and is removable from, the drawer **144A**. The drawer front panel **144B** is mounted to the drawer **144A** and the drawer handle **144C** is mounted to the drawer front panel **144B**.

(89) With reference to FIGS. **6A-6C**, **7A-7D**, **8A-8F** and **9A-9E**, the lower assembly **200** includes an outer enclosure **300**, an inner enclosure **400** and the fire box **204**. The lower assembly **200** is shown in FIG. **6C** without the enclosure **116**, the drawer assembly **144**, and the ceramic cooking stones **62** to illuminate the interior of lower assembly **200**. With specific reference to FIG. **6C**, a pair of stone main supports **208** are mounted to the fire box **204** to support the ceramic cooking

stones **62**.

(90) With specific reference to FIGS. **7A-7C**, the outer enclosure **300** includes an outer enclosure back panel **302** and first and second outer enclosure side panels **304**, **306**. The outer enclosure back panel **302** and first and second outer enclosure side panels **304**, **306** are composed from stainless steel held together using appropriate fasteners (not shown). The first and second outer enclosure side panels **304**, **306** include a plurality of first and second outer enclosure side panel flanges **304A**, **306A** and a plurality of side panel vents **304B**, **306C**. The first and second outer enclosure side panel flanges **304A**, **306A** includes a plurality of apertures for receiving the fasteners. As explained in further detail below, the side panel vents **304B**, **306B** allow external ambient air to enter the internal cavity **202** of the lower assembly **200**.

(91) The outer enclosure back panel **302** includes an opening **302A**. Power is provided to the oven **10** via a transformer **312**. The transformer **312** is mounted to a transformer cover **314** which is mounted to the outer enclosure back panel **302** adjacent the opening **302A**. An outer back cover plug cover **308** is also mounted to the outer enclosure back panel **302** to provide protection to the internal cavity **202**. A cable grommet **310** is positioned within an aperture in the outer back cover plug cover **308**. The cable grommet **310** allows passage of a transformer power cord **316** (only a portion of which is shown).

(92) With specific reference to FIGS. **8A-8H**, the inner enclosure **400** includes heat shield **402**, a fire box support structure **404** and a crumb collection assembly **406**. The heat shield **402** provides thermal isolation between the fire box **204** and the rest of the lower assembly **200**. The fire box **204** is supported by the fire box support structure **404**. The inner enclosure **400** is shown without the heat shield **402** in FIGS. **8B** and **8C**.

(93) The crumb collection assembly **406** allows crumbs funneled from the top of the cooking stones **62** (via opening **62A** in one of the cooking stones **62**) into a crumb collection slot **406A** down a crumb collection chute **406B** and into a removable crumb collection drawer **406C**. The removable crumb collection drawer **406D** slides along a pair of crumb collection drawer rails **406D** mounted to a bottom surface of the fire box support structure **404** (see FIG. **8H**). In the illustrated embodiment, the inner enclosure **400** includes a pair of spaced apart bottom support brackets **408** mounted across the bottom of the inner enclosure **400**. A pair of spaced apart storage drawer brackets **410** are mounted to the bottom support brackets **408**. A pair of storage drawer slides **412** are mounted to the bottom bracket **408**. The drawer assembly **144** is mounted or connected to the drawer slides **412** to allow the drawer assembly **144** to be moved from open and closed positions.

(94) In the illustrated embodiment, the heat shield **402** includes an upper vertical portion **402A** and a lower horizontal portion **402B**. The upper vertical portion **402A** extends from a back edge of the lower horizontal portion **402B** at a right angle. With specific reference to FIGS. **8D-8F**, the heat shield **402** includes a lower pan **402C**, an insulation panel **402D**, and an upper pan **402E**. The lower pan **402C** and the upper pan **402E** each have an inner cavity **202** and fit together to fully enclose the insulation panel **402D**. In one embodiment of the present invention, the lower pan **402C** and the upper pan **402E** composed of stainless steel and are fastened together using a plurality of fasteners (not shown). A heat shield front panel **402F** is mounted to the fire box support structure **404**.

(95) As shown in FIGS. **8A-8B** and **8G-8H**, the fire box support structure **404** includes a number of fire box support brackets **414** mounted to an interior of the fire box support structure **404**. The fire box **204** rests within the fire box support structure **404** and is supported by the fire box support brackets **414**.

(96) In the illustrated embodiment, the first box support structure **404** includes a fire box support structure top panel **416** and a pair of fire box support structure side panels **418**. In the illustrated embodiment, the fire box support structure top panel and side panels **416**, **418** are composed from stainless steel and fastened together using a plurality of fasteners (not shown).

(97) As shown, in the illustrated embodiment the fire box support structure top panel **416** includes a plurality of vents **416A** located on opposite sides. Further, each fire box support structure side panel

418 includes a plurality of lower vents **418A** and a plurality of upper vents **418B**.

(98) With reference to FIGS. **9A-9E**, the illustrated embodiment the fire box **204** is composed from stainless steel and includes a fire box top panel **210**, a fire box rear panel **212**, a fire box front panel **214**, opposing fire box side panels **216**, and a bottom panel **218**. The fire box top panel **210** includes a plurality of fire box top panel vents **210A** and the fire box rear panel **212** includes a plurality of fire box rear panel vents **212A**. Additional, each of the side panels **216** includes a plurality of fire box side panel vents **216A**.

(99) Air enters the oven **10** from below the oven **10** and/or through the vents **306B** in the side panels **316** of the outer enclosure **300** and the lower and upper vents **418A**, **418B** of the fire box support structure side panels **418**. Then, air enters the fire box **204** via vents **212A**. The first and second airflow pathways **110**, **112** start within the fire box **204** and exit the fire box **204** through the vents **216A** in the fire box side panels **216**. Air may also enter the oven **10** via gaps or vents located around the control panel **142A**.

(100) The first and second pathways **110**, **112** proceeds through the vents **216A** at either side of the top panel **216** of the fire box **204**. As shown in FIGS. **2A-2B**, the first pathway **110** traverses between the outer dome **106** and the inner dome **108**. The first pathway **110** ends at the heating vents **114** in the inner walls **108A** of the inner dome **108**. The second pathway **112** also traverses between the outer dome **106** and the inner dome **108** and terminates at one of the upper and lower vents **118J**, **118O** in the outer dome top panel **118**.

(101) As discussed previously, gas is provided to the oven **10** via the swiveling elbow connector **116D**. With reference to FIG. **10**, gas is controllably provided to the first and second gas burners **206A**, **206B** using the first and second burner knob assemblies **20**, **22** and a valve assembly **320**. The valve assembly **320** includes a gas manifold **322** and first and second manifold burner valves **324**, **326**. The gas manifold **322** is connected to the swiveling elbow connector **116D** via a first hose adapter **328**, a first corrugated hose **330** and a second hose adapter **332**. The first and second burner knob assemblies **20**, **22** are mounted to respective stems **324A**, **326A** of the first and second manifold burner valves **324**, **326**. The gas manifold **322** is mounted to the fire box **204** via a pair of gas mounting brackets **338**.

(102) In the illustrated embodiment, the main gas burner **206A** is an H-shaped burner having rated at **18,000** BTU. A burner air shutter **340** is connected to an input end of the H-shaped gas burner **206A**. A burner orifice connector **342** is connected to the burner air shutter **340**. A second corrugated hose (not shown) is between an output end **324B** of the first manifold burner valve **324** and the burner orifice connector **342** via a third hose connector **334**.

(103) A first igniter assembly **344** is coupled to the first burner knob assembly **20**. The first burner knob assembly **20** controllably supplies gas to the first gas burner **206A** which then may be ignited by the first igniter assembly **344** using the first ignition button **24**.

(104) With reference to FIGS. **10** and **11**, in the illustrated embodiment a second burner air shutter is connected to an end of the second burner **206B**. A second burner orifice connector **348** is connected to the second burner air shutter **346**. A third corrugated hose (not shown) is connected between an output end **326B** of the second manifold burner valve **326A** (via a fourth hose adapter **336**) and the second burner orifice connector **348**. A second igniter assembly **350** is coupled to the second burner knob assembly **22**. The second burner knob assembly **22** controllably supplies gas to the second gas burner **206B** which then may be ignited by the second igniter assembly **350** using the second ignition button **26**.

(105) The invention has been described in an illustrative manner, and it is be understood that the terminology which has been used is intended to be in the nature of words of description rather than of limitation. It is now apparent to those skilled in the art that many modifications and variations of the present invention are possible in light of the above teachings. It is, therefore, to be understood that the invention may be practiced otherwise than as specifically described.

Claims

1. An oven, comprising: an upper assembly having an upper portion and a bottom, and the upper assembly forming a cooking chamber and having a front side with an open entry to the cooking chamber; a lower assembly located below the upper assembly with the lower assembly having an internal cavity; a fire box located within the internal cavity of the lower assembly and having a first heat source; a heat absorbing and emanating device located between the upper and lower assemblies and being configured to absorb heat from the heat source and emanate heat into the cooking chamber of the upper assembly; and a see-through slide indicator having an indicator knob mounted within a single opening in the lower assembly, the indicator knob moveable within the single opening between an open position and a closed position with the firebox viewable within the internal cavity through the single opening when the indicator knob is in the open position.
2. An oven, comprising: an upper assembly having an upper portion and a bottom, and the upper assembly forming a cooking chamber and having a front side with an open entry to the cooking chamber; a lower assembly located below the upper assembly with the lower assembly having an internal cavity; a fire box located within the internal cavity of the lower assembly and having a first heat source; and, a heat absorbing and emanating device located between the upper and lower assemblies and being configured to absorb heat from the heat source and emanate heat into the cooking chamber of the upper assembly, wherein the upper assembly includes an outer dome and an inner dome with the inner dome being located within the outer dome and forming first and second airflow pathways disposed between the inner and outer domes, each of the first and second airflow pathways having an open end located at the bottom of the upper assembly and being open to the internal cavity of the lower assembly with the first airflow pathway having a second end terminating at a vent located on an inner wall of the inner dome and configured to allow heated air from the heat source to enter the cooking chamber, and the second airflow pathway having a second end terminating in the upper portion of the upper assembly, wherein the inner dome of the upper assembly further includes an inner dome upper portion coupled to the inner wall of the inner dome, with the inner dome upper portion having a top surface and a plurality of side panels extending from the top surface, and wherein the top surface is closed such that the heated air within the cooking chamber of the inner dome is unable to exit through the top surface of the inner dome upper portion.
3. An oven, as set forth in claim 2, wherein the heat absorbing and emanating device includes at least one ceramic stone.
4. An oven, as set forth in claim 2, further including a second heat source located within the upper assembly for heating the cooking chamber.
5. An oven, as set forth in claim 4, further including a heat guard located between the second heat source and the cooking chamber.
6. An oven, as set forth in claim 2, wherein the upper assembly includes at least one exhaust vent located within the upper portion of the upper assembly near the second end of the second airflow path for allowing heated gas to exit the oven.
7. An oven, as set forth in claim 2, further including a thermometer for sensing a temperature associated with the cooking chamber.
8. An oven, as set forth in claim 7, wherein the thermometer includes an indicator for providing a visual indication of the temperature and a stem extending from the indicator into the cooking chamber.
9. An oven, as set forth in claim 2, further including at least one removable door rotatably coupled to the upper assembly within the open entry.
10. An oven, as set forth in claim 2, wherein the inner dome includes at least one insulation pad configured to retain heat within the cooking chamber.

11. An oven, as set forth in claim 2, wherein the outer dome forms an exterior of the upper assembly, the inner dome forms at least a portion of the cooking chamber, the first and second airflow pathways being located between the inner and outer domes.
12. An oven, as set forth in claim 11, wherein the inner dome includes at least one lower portion side panel, wherein the first and second airflow pathways are located between the exterior of the upper assembly and the at least one lower portion side panel.
13. An oven, as set forth in claim 11, wherein the inner dome includes first and second lower portion side panels located on opposite sides of the inner dome, wherein the first and second airflow pathways are located between the exterior of the upper assembly and each lower portion side panel.
14. An oven, as set forth in claim 13, wherein the heating vent is located within each of the side panels.
15. An oven, as set forth in claim 11, wherein the outer dome includes an outer dome top panel, the outer dome top panel having a rectangular shape defined by front, back and first and second side panels, the outer dome top panel further including a rectangular top surface and a plurality of sloping walls, each sloping wall extending from an edge of the rectangular top surface to a respective one of the front, back and side panels.
16. An oven, as set forth in claim 15, including a first exhaust vent located in one of the sloping walls, the second airflow path ending at the first exhaust vent.
17. An oven, as set forth in claim 16, the outer dome top panel including at least one flange facing downward and extending from a respective one of the front, back and side panels, the at least one flange including a second exhaust vent.
18. An oven, as set forth in claim 16, wherein one of the plurality of side panels includes an exhaust opening to permit heated air to exit the cooking chamber and route the heated air towards the first exhaust vent located in one of the sloping walls of the outer dome.
19. An oven, as set forth in claim 15, wherein the upper assembly further includes a second heat absorbing and emanating device within the second airflow pathway and located adjacent a top portion of the cooking chamber.
20. An oven, as set forth in claim 2, wherein one of the plurality of side panels includes an exhaust opening to permit heated air to exit the cooking chamber.
21. An oven, as set forth in claim 2, wherein the top surface includes an opening, and further comprising a ceramic cooking stone resting upon the top surface such that the ceramic cooking stone is exposed to the cooking chamber.
22. An oven, comprising: an upper assembly having an upper portion and a bottom, and the upper assembly forming a cooking chamber and having a front side with an open entry to the cooking chamber; a lower assembly located below the upper assembly with the lower assembly having an internal cavity; a fire box located within the internal cavity of the lower assembly and having a first heat source; a second heat source located within the upper assembly for heating the cooking chamber; a heat guard located between the second heat source and the cooking chamber; a thermometer coupled to the lower assembly for sensing a temperature associated with the cooking chamber; at least one removable door rotatably coupled to the upper assembly within the open entry; and, a heat absorbing and emanating device located between the upper and lower assemblies and being configured to absorb heat from the heat source and emanate heat into the cooking chamber of the upper assembly, wherein the upper assembly includes an outer dome and an inner dome with the inner dome being located within the outer dome and forming first and second airflow pathways disposed between the inner and outer domes, each of the first and second airflow pathways having an open end located at the bottom of the upper assembly and being open to the internal cavity of the lower assembly with the first airflow pathway having a second end terminating at a vent located on an inner wall of the inner dome and configured to allow heated air from the heat source to enter the cooking chamber, and the second airflow pathway having a

second end terminating in the upper portion of the upper assembly, wherein the upper assembly further includes at least one exhaust vent located within the upper portion of the upper assembly in the outer dome near the second end of the second airflow path for allowing heated gas to exit the oven, wherein the outer dome forms an exterior of the upper assembly, the inner dome forms at least a portion of the cooking chamber, the first and second airflow pathways being located between the inner and outer domes, wherein the inner dome of the upper assembly further includes an inner dome upper portion coupled to the inner wall of the inner dome, with the inner dome upper portion having a top surface and a plurality of side panels extending from the top surface, and wherein the top surface is closed such that the heated air within the cooking chamber of the inner dome is unable to exit through the top surface of the inner dome upper portion.
